

ENTANGLED WERNER STATES UNDETECTED BY EVERY ENTANGLEMENT TESTER

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STATUS

This is a candidate counterexample, subject to human review, to the main open question of Jivulescu, Lancien, and Nechita [1]. Without the index permutation allowed in their positive theorem, the family of entanglement testers is not complete. In fact, a whole interval of entangled Werner states is invisible to every possible tailored pair of testers.

SOURCE QUESTION

For a multipartite state ρ , the source applies contractions $\mathcal{E}_i : S_1^{d_i} \rightarrow \ell_2^{n_i}$ and tests whether

$$\|(\mathcal{E}_1 \otimes \cdots \otimes \mathcal{E}_m)(\rho)\|_{\ell_2^{n_1} \otimes \pi \cdots \otimes \pi \ell_2^{n_m}} > 1.$$

It asks whether every entangled state is detected in this way, without first permuting indices. The source notes that in the bipartite case this asks whether every entangled state has a norm-one witness factoring through ℓ_2 [1, Section 13, p. 35].

us to prove two recent conjectures. We then show that every entangled bipartite mixed quantum state can be detected, after index permutation, by a pair of specifically tailored testers, and that the realignment tester can detect any multipartite pure entangled state.

The main question that remains unanswered at this point is whether our family of entanglement criteria is complete, without allowing for permutation of indices before applying the testers. Concretely, we would like to know if, for any entangled state ρ on $\mathbb{C}^{d_1} \otimes \cdots \otimes \mathbb{C}^{d_m}$, there exist testers $\mathcal{E}_i : S_1^{d_i} \rightarrow \ell_2^{n_i}$, $1 \leq i \leq m$, such that $\|\mathcal{E}_1 \otimes \cdots \otimes \mathcal{E}_m(\rho)\|_{\ell_2^{n_1} \otimes \pi \cdots \otimes \pi \ell_2^{n_m}} > 1$. Even for bipartite states we are only able to show a weaker version of this statement (see Section 11). Note that, in the bipartite case, this problem can be seen as a *factorization through ℓ_2* problem. Indeed, given an entangled state ρ on $\mathbb{C}^d \otimes \mathbb{C}^d$, there exists by definition an operator $\mathcal{T} : S_1^d \rightarrow S_\infty^d$ with norm at most 1 witnessing its entanglement. And in order to exhibit testers detecting its entanglement, we would need to find operators $\mathcal{E}, \mathcal{F} : S_1^d \rightarrow \ell_2^d$ with norms at most 1 such that $\mathcal{T} = \mathcal{E}^* \mathcal{F}$. It can be shown that not every $\mathcal{T} : S_1^d \rightarrow S_\infty^d$ can be factorized in this way with constant 1 [Aub20]. However, this does not tell us if there exist entangled states that do not have any factorizable entanglement witness.

In a different direction, it would be worth investigating further the performance of our entanglement criteria in the multipartite setting. Indeed, the only quantitative results that we establish in

PROOF INTUITION

The source already identifies the right geometric object: tester witnesses are maps $S_1^d \rightarrow S_\infty^d$ with Hilbert-factorization norm at most one. Werner states are invariant under $U \otimes U$. Averaging any such witness over this symmetry preserves both its value on the state and its factorization bound, and leaves only a two-parameter witness $aI + bF$.

The decisive calculation is exact. For the map represented by $aI + bF$,

$$\gamma_2(T_{a,b}) = \max \left\{ |b|, \left| a + \frac{b}{d} \right| + \left(1 - \frac{1}{d} \right) |b| \right\}.$$

The upper bound splits a matrix into its scalar and traceless parts and sends the latter through Hilbert–Schmidt space. For the lower bound, an off-diagonal matrix unit sees $|b|$, while restriction to diagonal matrices produces $aJ + bI$; its nuclear norm divided by d gives the other term. The remaining two-variable optimization shows that no tailored tester beats ordinary realignment on a Werner state.

FACTORABLE WITNESSES

Use the complex bilinear trace duality between S_1^d and S_∞^d . For $W \in \mathcal{M}_d \otimes \mathcal{M}_d$, let $T_W : S_1^d \rightarrow S_\infty^d$ be determined by

$$\text{Tr}(T_W(X)Y) = \text{Tr}(W(X \otimes Y)).$$

Changing the tensor-to-map convention can insert a transpose, which is an isometry on both Schatten classes and does not affect any norm below. Put

$$\gamma_2(T) = \inf \{ \|A\| \|B\| : T = BA, A : S_1^d \rightarrow H, B : H \rightarrow S_\infty^d, H \text{ Hilbert} \}.$$

For a state ρ , define its all-tester envelope

$$\alpha(\rho) = \sup_{\mathcal{E}, \mathcal{F}} \|(\mathcal{E} \otimes \mathcal{F})(\rho)\|_\pi,$$

where the supremum is over contractions from the two local S_1 spaces to arbitrary finite-dimensional Hilbert spaces. Requiring the maps to have norm exactly one gives the same detection question.

Lemma 1 (Tester duality). *For every bipartite state ρ ,*

$$\alpha(\rho) \leq \sup_{\gamma_2(T_W) \leq 1} |\text{Tr}(W\rho)|.$$

Proof. Fix contractions

$$\mathcal{E} : S_1^d \rightarrow H, \quad \mathcal{F} : S_1^d \rightarrow K.$$

By projective/injective duality on $H \otimes K$, each functional used to norm $(\mathcal{E} \otimes \mathcal{F})(\rho)$ is a bounded bilinear form v on $H \times K$ of norm at most one. By the Riesz representation theorem, v is represented by a contraction between the two Hilbert spaces. Hence

$$(X, Y) \mapsto v(\mathcal{E}X, \mathcal{F}Y)$$

has associated map $S_1^d \rightarrow S_\infty^d$ factoring through a Hilbert space with product of norms at most one. Its representing tensor is some W with $\gamma_2(T_W) \leq 1$. Taking the successive suprema proves the claim. $\square$

THE EXACT INVARIANT FACTORIZATION NORM

Let $F(x \otimes y) = y \otimes x$ be the flip and put $W_{a,b} = aI + bF$. Then, up to the harmless transpose mentioned above,

$$T_{a,b}(X) = a \operatorname{Tr}(X)I + bX.$$

Lemma 2. *For every $a, b \in \mathbb{C}$ and $d \geq 2$,*

$$(1) \quad \gamma_2(T_{a,b}) = \max \left\{ |b|, \left| a + \frac{b}{d} \right| + \left(1 - \frac{1}{d} \right) |b| \right\}.$$

Proof. Write

$$A = a + \frac{b}{d}, \quad c = 1 - \frac{1}{d}, \quad X_0 = X - \frac{\operatorname{Tr} X}{d} I.$$

For $r, s > 0$, factor $T_{a,b} = G_{r,s}E_{r,s}$ through $\mathbb{C} \oplus_2 S_2^{d,0}$ by

$$E_{r,s}(X) = (r \operatorname{Tr} X, sX_0), \quad G_{r,s}(z, Z) = \frac{A}{r} zI + \frac{b}{s} Z.$$

The extreme points of the complex S_1 unit ball are rank-one matrices uv^* with $\|u\| = \|v\| = 1$. If $t = \operatorname{Tr}(uv^*)$, then $\|(uv^*)_0\|_2^2 = 1 - |t|^2/d$. It follows, and by duality the same calculation applies to $G_{r,s}$, that

$$(2) \quad \|E_{r,s}\|^2 = \max\{s^2, r^2 + cs^2\},$$

$$(3) \quad \|G_{r,s}\|^2 = \max \left\{ \frac{|b|^2}{s^2}, \frac{|A|^2}{r^2} + c \frac{|b|^2}{s^2} \right\}.$$

If $b = 0$, the scalar factorization gives $\gamma_2(T_{a,0}) \leq |a|$. Assume $b \neq 0$ and set $B = |b|$, $C = |A|$, and $s = 1$. If $C \leq B/d$, choose r^2 in the nonempty interval

$$d(C/B)^2 \leq r^2 \leq 1/d.$$

Equations (2)–(3) give $\|E_{r,1}\| \|G_{r,1}\| = B$. If $C \geq B/d$, choose $r^2 = C/B$; the product is $C + cB$. This proves the upper bound in (1).

For the reverse bound, $\gamma_2(T) \geq \|T\|$. Applying $T_{a,b}$ to an off-diagonal matrix unit gives $\gamma_2(T_{a,b}) \geq |b|$. Next restrict the domain to diagonal matrices and project the range onto its diagonal. These maps have norm one, so the resulting operator

$$M = aJ + bI : \ell_1^d \longrightarrow \ell_\infty^d$$

has $\gamma_2(M) \leq \gamma_2(T_{a,b})$. If $M_{ij} = \langle x_j, y_i \rangle$ is any Hilbert factorization, then

$$\|M\|_* \leq \left(\sum_j \|x_j\|^2 \right)^{1/2} \left(\sum_i \|y_i\|^2 \right)^{1/2} \leq d \|A_0\| \|B_0\|,$$

where $A_0 : \ell_1^d \rightarrow H$ and $B_0 : H \rightarrow \ell_\infty^d$ are the two factors. Thus $\gamma_2(M) \geq \|M\|_*/d$. The eigenvalues (and singular values in modulus) of $aJ + bI$ are dA once and b with multiplicity $d - 1$. Consequently

$$\gamma_2(T_{a,b}) \geq \frac{\|M\|_*}{d} = |A| + c|b|.$$

Together with the off-diagonal bound, this proves equality. $\square$

WERNER STATES AND THE NEGATIVE ANSWER

For $0 \leq \mu \leq 1$, the Werner state [2, 1] is

$$(4) \quad \sigma_\mu = \mu \frac{I + F}{d(d+1)} + (1 - \mu) \frac{I - F}{d(d-1)}.$$

It is positive, has trace one, is invariant under $U \otimes U$, and satisfies

$$(5) \quad f_\mu := \text{Tr}(F\sigma_\mu) = 2\mu - 1.$$

Theorem 1 (Exact all-tester envelope). *For every $d \geq 2$ and $0 \leq \mu \leq 1$,*

$$(6) \quad \alpha(\sigma_\mu) = 1 + 2 \left(\frac{1}{d} - \mu \right)_+.$$

Hence the realignment tester is optimal among all tailored pairs of entanglement testers on every Werner state.

Proof. The γ_2 unit ball is convex and invariant under local unitary conjugations. Given a witness W in that ball, average it over $W \mapsto (U \otimes U)W(U^* \otimes U^*)$. Its value on σ_μ is unchanged, and the average remains in the ball. Taking a Hermitian part also does not increase γ_2 . By the elementary two-fold Schur–Weyl commutant, the averaged Hermitian witness has the form $aI + bF$ with $a, b \in \mathbb{R}$.

Put $A = a + b/d$ and $c = 1 - 1/d$. Lemma 2 says that the feasible region is

$$|b| \leq 1, \quad |A| + c|b| \leq 1.$$

Using (5), the objective is

$$\text{Tr}((aI + bF)\sigma_\mu) = A + b(f_\mu - 1/d).$$

For fixed b it is maximal at $A = 1 - c|b|$. Maximizing over $-1 \leq b \leq 1$ gives

$$\begin{aligned} \sup_{\gamma_2(T_W) \leq 1} |\text{Tr}(W\sigma_\mu)| &= 1 + (|f_\mu - 1/d| - c)_+ \\ &= 1 + 2(1/d - \mu)_+. \end{aligned}$$

The absolute value causes no extra branch because $-1 \leq f_\mu \leq 1$. Lemma 1 proves the corresponding upper bound for α .

For the reverse bound, if $\mu \geq 1/d$, the one-dimensional trace testers $X \mapsto \text{Tr } X$ give output norm $\text{Tr } \sigma_\mu = 1$. If $\mu < 1/d$, the realignment tester gives

$$\|(\mathcal{R} \otimes \mathcal{R})(\sigma_\mu)\|_1 = \frac{d+2}{d} - 2\mu = 1 + 2(1/d - \mu),$$

by the direct calculation in [1, Section 9]. Thus equality holds in all cases, and realignment attains the nontrivial branch. $\square$

Corollary 1 (Counterexamples to completeness). *For every $d \geq 3$ and*

$$\frac{1}{d} \leq \mu < \frac{1}{2},$$

σ_μ is entangled, but no pair of entanglement testers detects it. In particular, on $\mathbb{C}^3 \otimes \mathbb{C}^3$,

$$\rho_* := \sigma_{1/3} = \frac{5I - 3F}{36}$$

is an explicit counterexample to the source question.

Proof. For positive P, Q , $\text{Tr}(F(P \otimes Q)) = \text{Tr}(PQ) \geq 0$. Hence every separable state has nonnegative flip expectation. Equation (5) is negative when $\mu < 1/2$, so σ_μ is entangled. On the other hand, Theorem 1 gives $\alpha(\sigma_\mu) = 1$ when $\mu \geq 1/d$. Since the tester criterion requires a strict value greater than one, no tester pair detects these states.

For $d = 3$, F has eigenvalue $+1$ with multiplicity six and -1 with multiplicity three. Thus ρ_* has eigenvalues $1/18$ (six times) and $2/9$ (three times), has trace one, and $\text{Tr}(F\rho_*) = -1/3$. This verifies all density-state and entanglement claims directly. $\square$

VERIFICATION NOTES

The proof uses the complex S_1 unit ball in Lemma 2; its rank-one extreme points are essential for the exact norm formulas (2)–(3). The diagonal lower bound is normalized by d , not $\sqrt{d}$: a factorization of a $d \times d$ matrix has at most d row vectors and d column vectors, so the product of their Frobenius norms is at most d times the factorization constant.

The supplementary script verifies positivity, trace, and flip expectation for ρ_* ; scans the invariant support problem for $d = 2, \dots, 10$; and checks the optimizing factorization parameters on 7000 random coefficient pairs. It is not used as proof.

NOVELTY CHECK AND SCOPE

A bounded local, arXiv, and web search on August 11, 2026 used the source id and title together with “entanglement testers”, “Werner”, “complete”, “factorization through ℓ_2 ”, and the exact open-question wording. It found no prior negative answer or exact all-tester Werner envelope. A 2021 workshop report still states the problem as open. The six works indexed by OpenAlex as citing the source through 2025 were checked at title/abstract or full-text citation context; accessible arXiv sources 2202.13993, 2402.09972, and 2503.21177 use the paper for adjacent tensor-norm or SIC/realignment results, not this completeness question. Novelty is plausible, not certified.

One bipartite counterexample answers the source’s multipartite universal question negatively. The result does not address whether testers are complete

for 2×2 states or after the source’s permitted index permutation; the latter is already positive in [1].

HUMAN REVIEW RECOMMENDATION

First verify Lemma 1’s normalization and the nuclear-norm lower bound in Lemma 2. Then check the one-line support optimization in Theorem 1. If those pass, the explicit 3×3 state is a complete negative answer and the full Werner-state optimality statement is an additional sharp result.

REFERENCES

- [1] M. A. Jivulescu, C. Lancien, and I. Nechita, *Multipartite entanglement detection via projective tensor norms*, Ann. Henri Poincaré **23** (2022), 3791–3838; arXiv:2010.06365.
- [2] R. F. Werner, *Quantum states with Einstein–Podolsky–Rosen correlations admitting a hidden-variable model*, Phys. Rev. A **40** (1989), 4277–4281.